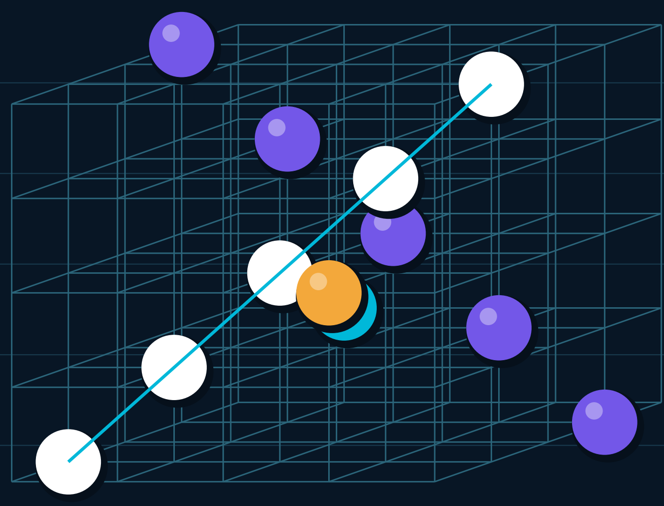
ADDITIONAL INFORMATION FOR JUDGES AND ORGANIZERS

3D CONNECT6

A spatial strategy game where every line becomes a new dimension



Created by Lidong Wang in collaboration with Codex (5.6)

Human-led product direction. AI-assisted engineering. Joint iteration and verification.

THINK BEYOND THE BOARD. CONNECT SIX ACROSS THREE DIMENSIONS.

Project dossier / Version 1.0 / 21 July 2026

01 Executive Brief

A complete, playable reinterpretation of Connect6 designed for strategic depth, visual clarity, and multiple forms of human-AI interaction.

1,000	13	4	48/48
PLAYABLE 3D CELLS	WIN DIRECTIONS	CORE PLAY EXPERIENCES	ENGINE TESTS PASSING

The Project in One Sentence

3D Connect6 transforms a familiar two-dimensional alignment game into a 10 x 10 x 10 spatial strategy environment, combining real-time multiplayer, local tactical AI, optional LLM opponents, training analysis, replay, and purpose-built 3D inspection tools.

The Design Question

Can a classic rule set remain understandable when its possibility space expands into three dimensions - without reducing the experience to a confusing cloud of pieces?

The Product Answer

Preserve simple Connect6 turns, then add camera-relative interaction, coordinate preview, transparency, occlusion awareness, slice inspection, replay, and AI guidance.

Core Rules

BOARD

The default board is 10 x 10 x 10. Coordinates use a right-handed X/Y/Z system.

TURN

Black places one opening stone. From the next turn onward, each player places two stones.

VICTORY

The first player to connect six stones along any orthogonal, face-diagonal, or space-diagonal line wins.

Why It Matters

- It turns a known abstract game into a new spatial reasoning challenge while keeping the win condition immediately recognizable.
- It is both a game and an AI experimentation surface: human vs AI, assisted training, AI vs AI observation, and online human competition.
- Its shared deterministic engine keeps browser play, server validation, replay, and testing aligned around one source of truth.

02 Game and Player Experience

The interface is designed to help players form a reliable mental model of a dense board while keeping turns deliberate and readable.

Four Ways to Engage

MODE	PURPOSE	DISTINCTIVE VALUE
Human vs AI	Play locally against the selected intelligence source.	Choose side or random assignment; local computation avoids network dependency.
Training	Explore positions and request tactical analysis.	Move explanations expose threats, extensions, blocks, and immediate wins.
AI vs AI	Observe two selected agents compete.	A spectator mode for comparing behavior without blocking the visual thread.
Online Multiplayer	Join a named room for a synchronized match.	Ready checks, random colors, a 90-second clock, reset consent, and reconnect-aware state.

Making Three Dimensions Legible

INSPECT

Orbit the scene, use axis labels, enable transparency, and hover to reveal potentially occluded stones.

SLICE

Open a 2D monitor for any X, Y, or Z layer to examine board cross-sections precisely.

REPLAY

Step backward and forward through move history without mutating the live game state.

Deliberate Placement

Players can enter coordinates, preview the target with a luminous marker, and confirm the move. Click and keyboard-assisted navigation provide additional interaction paths.

Readable Feedback

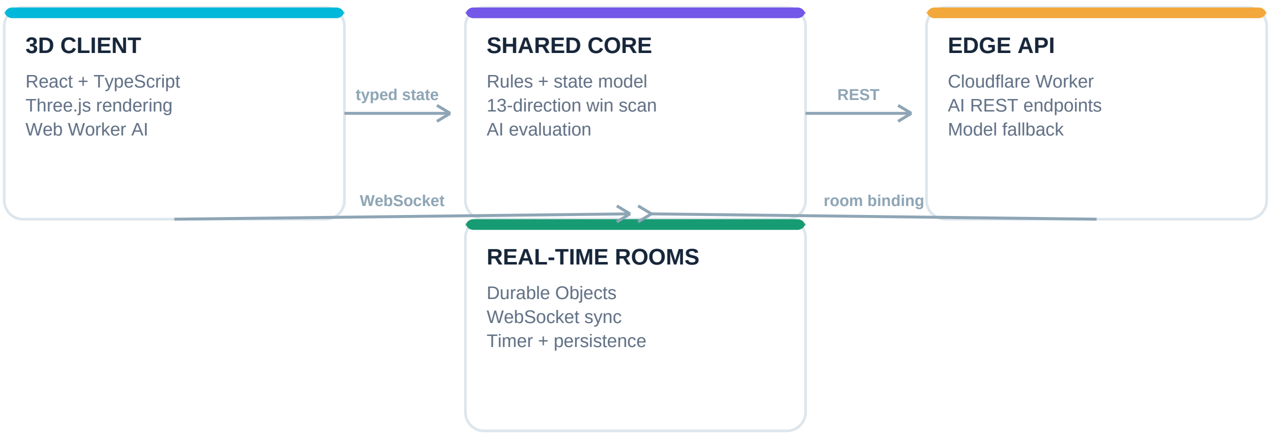
Current player, stone count, AI source, thinking state, winning line, timer, room readiness, and reset status are surfaced in the interface.

Experience Principle

The project does not hide the complexity of 3D play; it gives players multiple synchronized views of that complexity so they can reason about it.

03 Engineering Architecture

A TypeScript monorepo separates the interactive client, deterministic game core, and edge-hosted multiplayer/AI services.



Core Engineering Decisions

One Shared Engine

The rule engine has no browser or Node.js dependency. The same move validation, board indexing, serialization, and win logic can run in the client and Cloudflare Worker.

Constant-Time Win Scan

After each move, the engine scans both directions along 13 basis vectors. Runtime is O(13 x win length), effectively constant for the fixed six-stone target.

Authoritative Real-Time Rooms

Each room is mapped to a Cloudflare Durable Object. WebSockets carry typed messages while persisted snapshots support continuity and reconnect-aware behavior.

Responsive AI Execution

Local AI computation is moved into Web Workers so search and AI-vs-AI matches do not block Three.js rendering or user interaction.

Technology Stack

LAYER	TECHNOLOGIES	ROLE
Interface	React 18, TypeScript, Vite, Tailwind CSS	Application state, UI, styling, production bundling
3D	Three.js, React Three Fiber, Drei, mesh BVH	Scene graph, camera, lighting, interaction, spatial acceleration
Game Core	Pure TypeScript shared package	Rules, state, AI, serialization, deterministic tests
Edge Backend	Cloudflare Workers, Durable Objects, WebSockets	Rooms, validation, persistence, clocks, AI endpoints
Quality	Vitest, TypeScript compiler, Vite build	Behavioral tests, static checking, production compilation

04 Intelligence Layer

The project combines deterministic tactical safeguards with optional model reasoning, then validates every proposed action before it reaches the board.

POSITION	→	TACTICS	→	SEARCH	→	MODEL	→	VALIDATE
typed board snapshot		wins, blocks, threats		scoring + minimax		local or selected LLM		legal moves + fallback

Local Tactical AI

- Scores candidate moves by offensive potential, defensive urgency, open-line growth, centrality, and learned position memory.
- Prioritizes immediate wins and forced blocks, examines multi-threat patterns, and uses forward search/minimax for deeper choices.
- Runs off the main thread, preserving camera and board responsiveness during computation.

Optional LLM Opponents

Structured Spatial Context

The server describes board size, current color, stones remaining, critical threats, and 3D direction rules. Supported selections in the current code include Qwen, DeepSeek, and GLM variants.

Guardrails and Continuity

Model output is parsed into coordinates, rejected if invalid, and completed or replaced by local AI when needed. External failure therefore degrades capability rather than breaking the game loop.

Learning and Explanation

MEMORY

Completed games update a lightweight position-to-move record, allowing repeated successful patterns to influence future scoring.

ANALYSIS

Training mode explains tactical intent in plain language and can request deeper model-based interpretation.

COMPARISON

AI-vs-AI mode lets observers compare different local or model-driven styles on the same rule system.

Why the Hybrid Approach

A language model can contribute strategic interpretation and variety, but game legality and immediate tactical safety should remain deterministic. The architecture gives each method the role it performs best.

AI ASSISTANCE IS OPTIONAL AT PLAY TIME: THE LOCAL ENGINE REMAINS A COMPLETE FALLBACK.

05 Challenges, Trade-offs, and Verification

The difficult parts were not only implementing the rules, but also making spatial state visible, keeping AI work responsive, and synchronizing competitive play.

CHALLENGE	RESPONSE	RESULT
Occlusion	Camera-aware transparency, hover reveal, and orthogonal slice views.	Players can inspect both the whole volume and exact cross-sections.
3D victory logic	Represent 13 positive basis directions and scan each in both signs.	One compact algorithm covers axes, face diagonals, and space diagonals.
AI frame drops	Move local AI work into Web Workers and keep UI state lightweight.	Search can continue without blocking scene rendering.
Multiplayer consistency	Use an authoritative Durable Object room with typed WebSocket events and stored snapshots.	Moves, clocks, readiness, resets, and reconnect state share one authority.
Unreliable model output	Parse, validate, fill missing moves, and fall back to deterministic AI.	The board never trusts free-form model output directly.

Verification Snapshot - 21 July 2026

ENGINE TESTS

Vitest: 2 test files passed; 48 tests passed; 0 failures.

PRODUCTION BUILD

Shared TypeScript package and Vite client build completed successfully; 691 modules transformed.

SERVER CHECK

Cloudflare Worker TypeScript typecheck completed successfully with no errors.

Known Limitations and Honest Next Steps

- The production build reports one large JavaScript chunk (about 1.21 MB minified / 351 KB gzip). Route- or feature-level code splitting is the clearest performance follow-up.
- LLM latency and availability depend on external model endpoints. Local AI fallback protects play continuity, but model quality and response time can vary.
- The 3D interaction model has a learning curve. A guided first-game tutorial and usability testing with new players would improve onboarding evidence.
- This verification covers automated engine tests, compilation, and type safety. It does not claim a fresh live production uptime or cross-browser test in this document.

Evaluation Note

The strongest current evidence is the deterministic rule engine and breadth of integrated play modes. The most valuable next engineering investment is bundle optimization plus structured user testing of 3D comprehension.

06 Authorship and AI Collaboration Disclosure

This project was completed through a human-led collaboration between Lidong Wang and Codex (5.6). The disclosure below is included so judges and organizers can evaluate the work transparently.

Lidong Wang - Human Creator

Originated and owned the project concept; defined the product goal and rules; directed feature priorities and visual intent; evaluated behavior; provided iterative feedback; made final decisions; and accepted the completed result.

Codex (5.6) - AI Collaborator

Assisted with architecture exploration, implementation, refactoring, debugging, prompt design, performance work, tests, documentation, and preparation of this dossier under human direction.

Jointly Developed Areas

- Game architecture, interaction details, AI behavior, multiplayer workflows, performance optimizations, and iterative quality improvements were shaped through back-and-forth collaboration.
- AI-generated suggestions and code were reviewed through repository iteration, automated tests, builds, runtime-oriented fixes, and human acceptance rather than treated as automatically correct.
- The human creator retained authorship responsibility, project direction, and final judgment. Codex functioned as an engineering and creative collaborator, not an autonomous submitter.

TRANSPARENCY STATEMENT

The project intentionally acknowledges material AI assistance. Its contribution is part of the project's story: a human defined and steered an ambitious spatial game, while an AI collaborator accelerated implementation and helped turn repeated feedback into a tested software system.

Suggested Evaluation Path

1	LEARN	Read the three rules: one opening stone, two stones per later turn, six in any 3D line.
2	INSPECT	Rotate the board, enable transparency, and compare the 3D scene with an X/Y/Z slice.
3	PLAY	Try Human vs AI or Training mode, then inspect move explanations and replay.
4	COMPARE	Run AI vs AI or a multiplayer room to see the shared engine across interaction models.

Closing Note

3D Connect6 is an experiment in spatial strategy, hybrid intelligence, and transparent human-AI creation. Its central achievement is not simply adding another axis; it is building the tools that make that axis playable.